



DATA VISUALIZATION · FINAL INDIVIDUAL PROJECT · SUMMER 2026

Amsterdam Airbnb

Price, regulation, and what the averages hide

6,377 listings · 545,162 reviews · 22 neighbourhoods · Inside Airbnb, June 2026 (CC BY 4.0)

THE ARGUMENT

Amsterdam's Airbnb market does the opposite of what everyone assumes

The city caps most short-term rentals at 30 nights a year to protect housing from tourism pressure. Ten questions test what the listing data actually shows about price, hosts and that regulation. Four widely held beliefs turn out to be wrong:

–27%

professional multi-listing hosts
charge less, not more

–23%

Superhosts charge less than
regular hosts

flat

the centre price premium
vanishes under the median

23%

of hosts close their calendar
below the 30-night cap

Every headline claim here contradicts a common assumption about the market — and one of them contradicts the mean in the same dataset.

Each question below is multi-dimensional: it relates variables, compares groups, or tracks change across time or space. Each is answered with one explanatory figure.

One source, three files, and one cleaning decision that matters

listings.csv	10,369 rows × 90 columns — price, location, room type, host, availability	Inside Airbnb
reviews.csv	545,162 rows — one row per review, used as a booking proxy	Inside Airbnb
neighbourhoods.geojson	22 official Amsterdam neighbourhood boundaries	Inside Airbnb

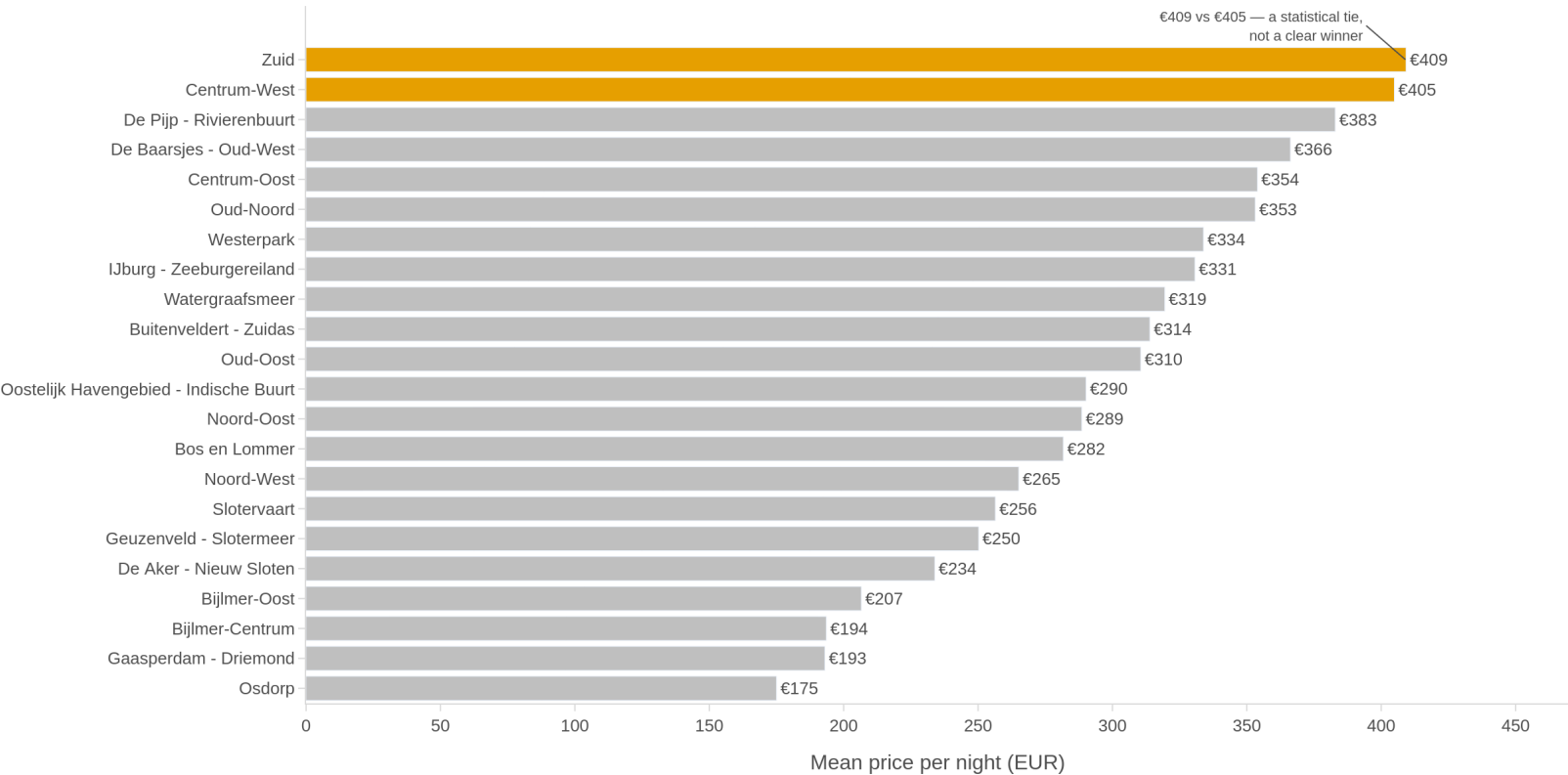
Attribute types and cleaning

- Numerical price, availability, review score · Categorical room type, host badge · Spatial coordinates and neighbourhood · Temporal 191 months of review dates
- Price parsed from a currency string; listings at €0 dropped; Superhost recoded from 't'/'f' to boolean
- 10,369 → 6,377 listings. The 38% drop is rows with no price or no neighbourhood — inactive or delisted entries that carry no rate
- Means are reported where the source chart used them, but the distribution is heavily right-skewed (median €291, 95th percentile €728, max €11,412) — Q8 shows what that costs

Design: Okabe–Ito CVD-safe palette, muted grey for context with one highlight colour, gridlines removed, titles state the finding.

QUESTION 1 . WHICH NEIGHBOURHOODS COMMAND THE BIGGEST PRICE PREMIUM?

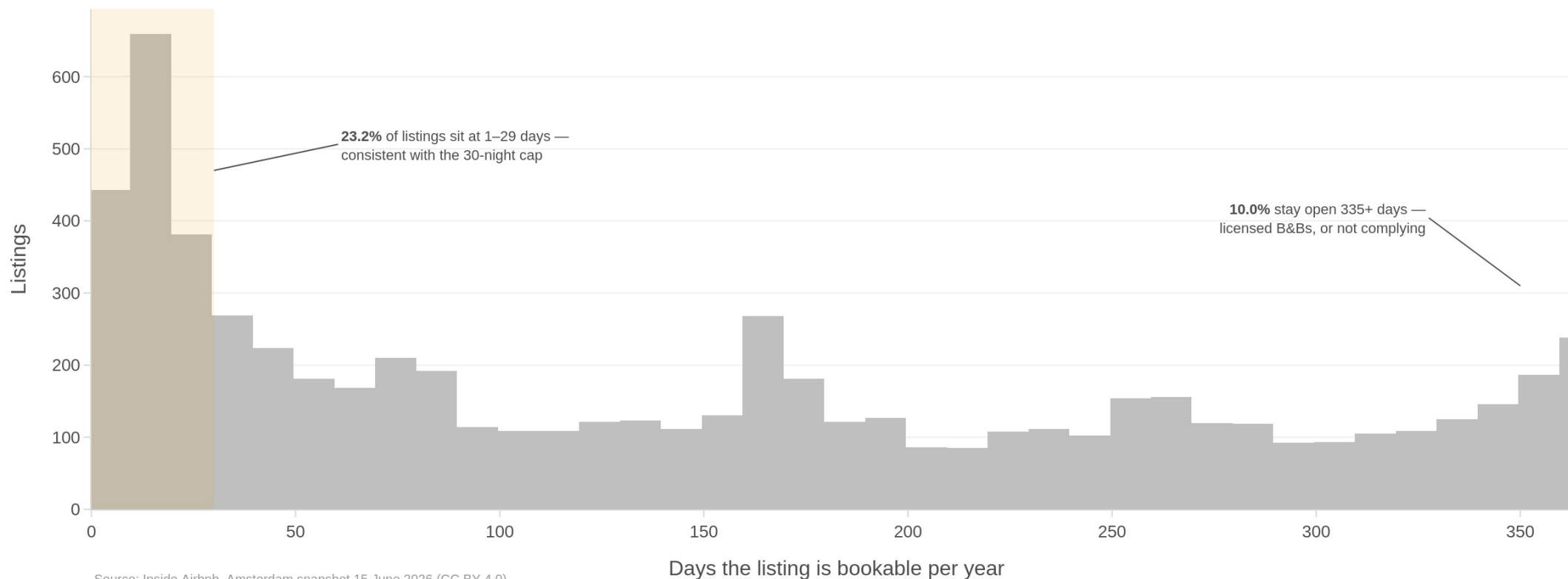
Zuid and Centrum-West lead — but they are separated by barely 1%



Mean nightly price runs from €175 in Osdorp to €409 in Zuid, a 2.3× spread across 22 neighbourhoods. But the top two, Zuid (€409.19) and Centrum-West (€404.87), differ by 1.1% — well inside the noise for samples of 435 and 851 listings. The honest reading is a leading group, not a single winner.

QUESTION 2 · DOES HOSTING ACTIVITY CLUSTER AROUND THE 30-NIGHT CAP?

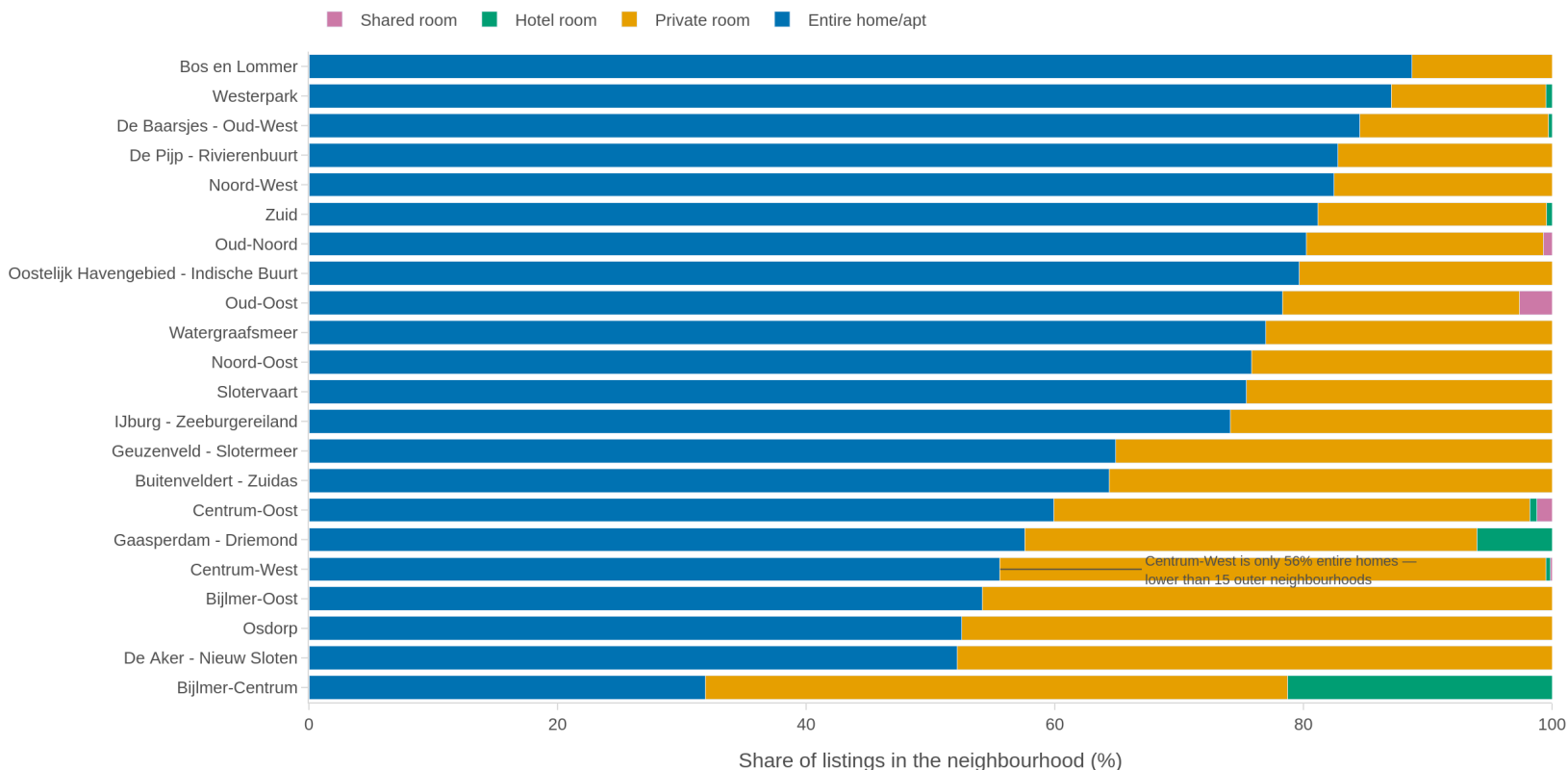
Most hosts close their calendar well below the cap — a tenth leave it wide open



23.2% of listings offer between 1 and 29 bookable days a year, consistent with hosts self-limiting to stay inside the 30-night rule. Only 0.5% sit at exactly 30, so there is no sharp spike at the threshold. At the other end, 10.0% stay open 335+ days — either licensed B&Bs and hotels that are exempt, or listings the cap is not reaching.

QUESTION 3 · HOW DOES THE ROOM-TYPE MIX VARY BY NEIGHBOURHOOD?

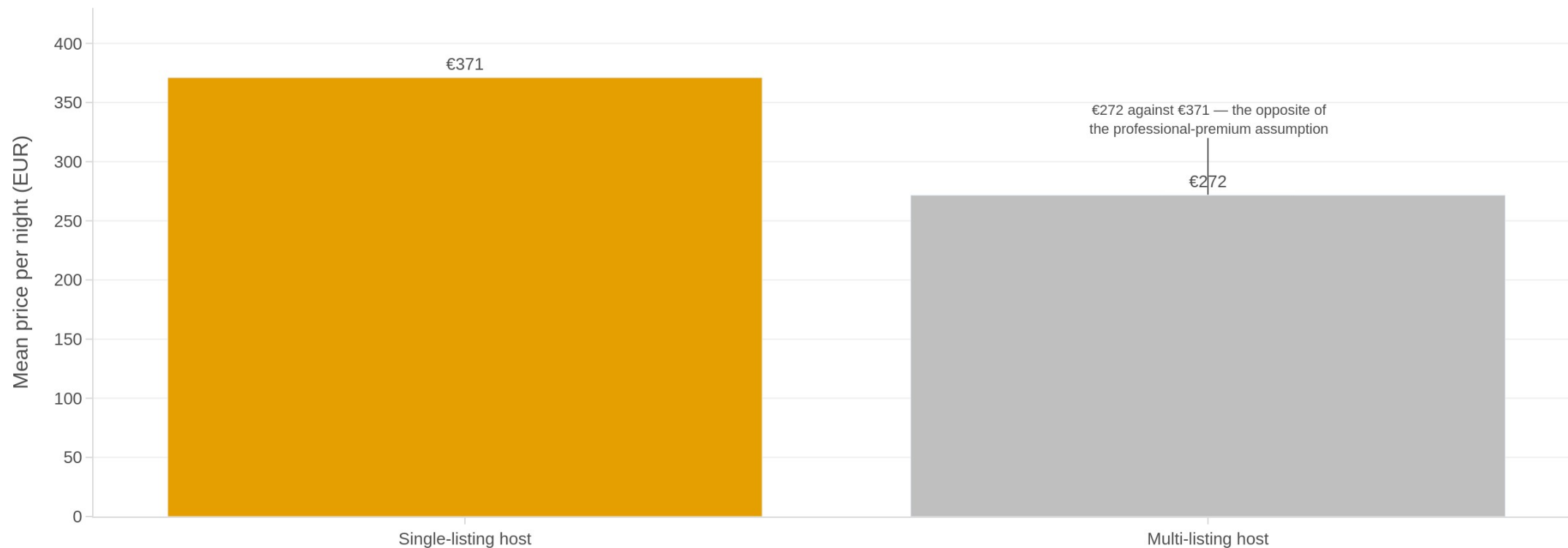
Entire homes dominate the ring, not the centre — the tourist core rents rooms



The intuition is backwards. Bos en Lommer (88.7%), Westerpark (87.1%) and De Baarsjes (84.5%) are almost entirely whole-home listings, while Centrum-West sits at 55.6% and Centrum-Oost at 59.9% — lower than fifteen outer neighbourhoods. The historic core still rents private rooms inside occupied homes; the residential ring is where whole flats have moved onto the platform.

QUESTION 4 · DO PROFESSIONAL MULTI-LISTING HOSTS PRICE DIFFERENTLY?

Professional multi-listing hosts charge 27% less, not more



Multi-listing hosts average €271.65 a night against €371.03 for single-listing hosts — the opposite of the professional-premium assumption. The likely mechanism is inventory mix rather than strategy: operators running several units tend to hold smaller, standardised flats, while a household letting its own single home is letting a larger property.

QUESTION 5 · HOW HAS BOOKING DEMAND MOVED OVER TIME?

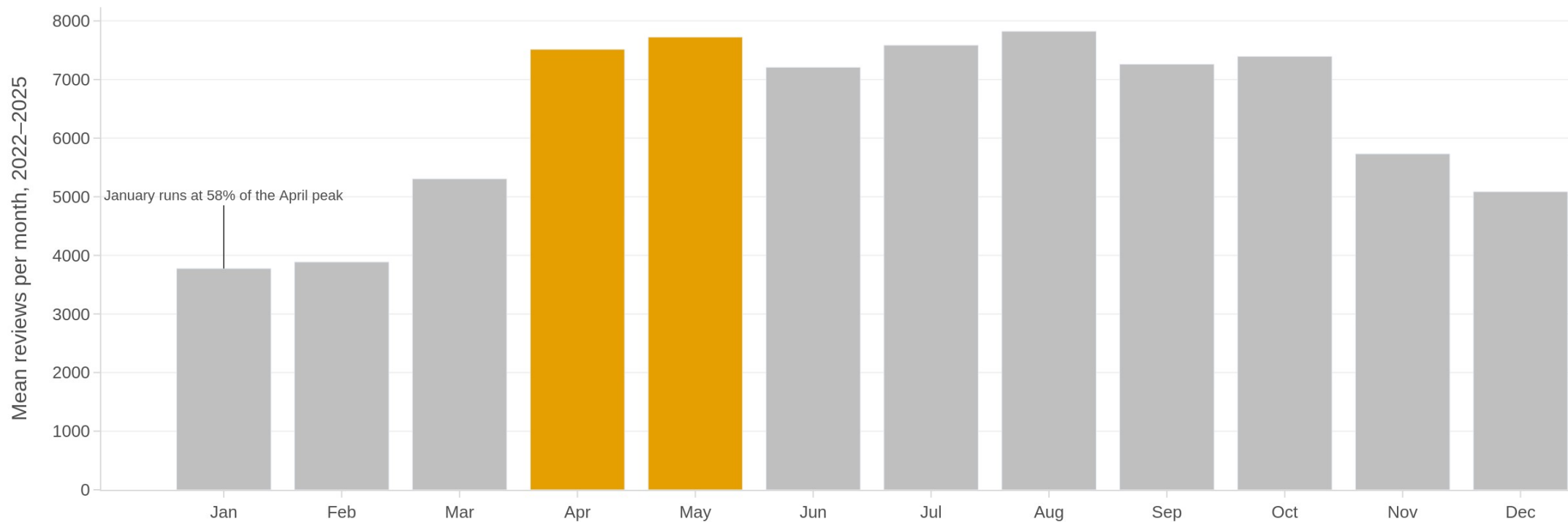
Demand collapsed in 2020 and came back nearly twice as strong



Review volume fell from 47,014 in 2019 to 17,074 in 2020, a 64% collapse. Recovery was complete by 2022 and has continued past the pre-pandemic baseline: 2025 closed at 89,421 reviews, 1.9× the 2019 level. Whatever restraint the 30-night cap imposes on individual hosts, it has not restrained aggregate demand.

QUESTION 6 · WHAT DOES THE SEASON ACTUALLY LOOK LIKE?

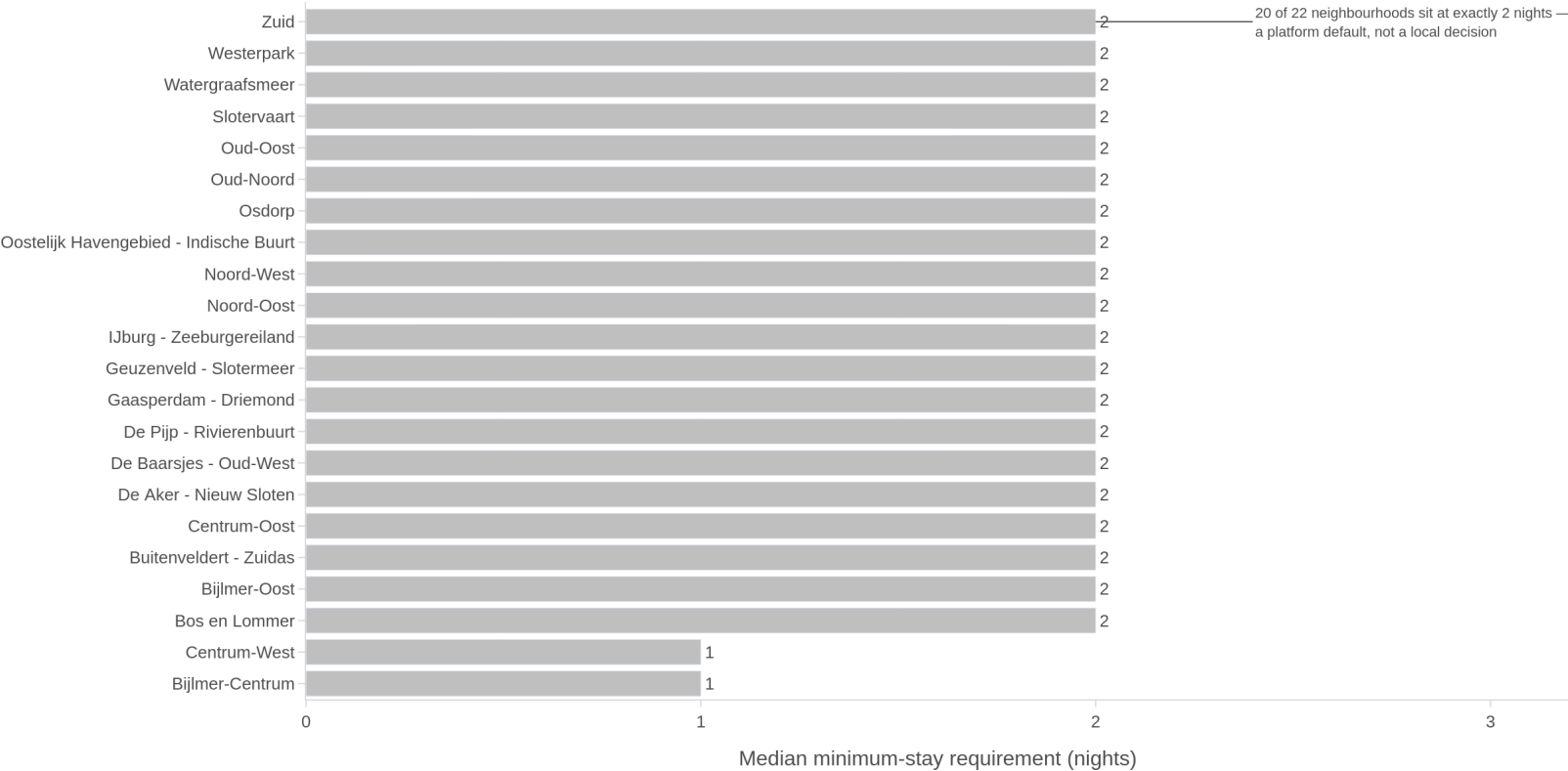
The season is a plateau from April to October, not a summer spike



Averaged across the four post-pandemic years, April and May are the busiest months, but June through October sit within 12% of them — a seven-month plateau rather than a peak. January runs at 58% of the April level. Amsterdam's tourism is far less seasonal than the summer-holiday model would predict, which matters for any policy timed around a peak.

QUESTION 7 · DO MINIMUM-STAY RULES DIFFER ACROSS THE CITY?

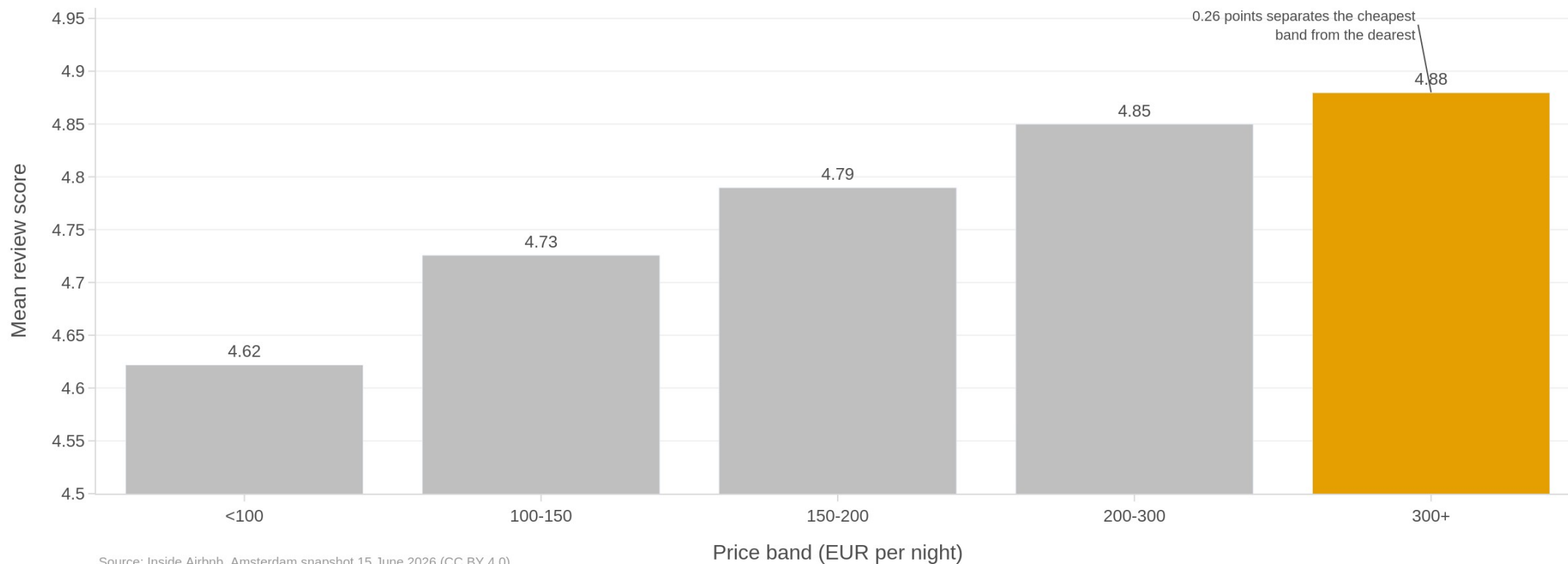
Minimum-stay rules are effectively uniform across the whole city



Twenty of twenty-two neighbourhoods sit at a median of exactly 2 nights; the remaining two sit at 1. There is no geographic variation to explain. This is a platform default being accepted almost universally rather than a local pricing decision — a negative result, and worth reporting as one.

QUESTION 8 · DO CHEAPER LISTINGS SCORE WORSE?

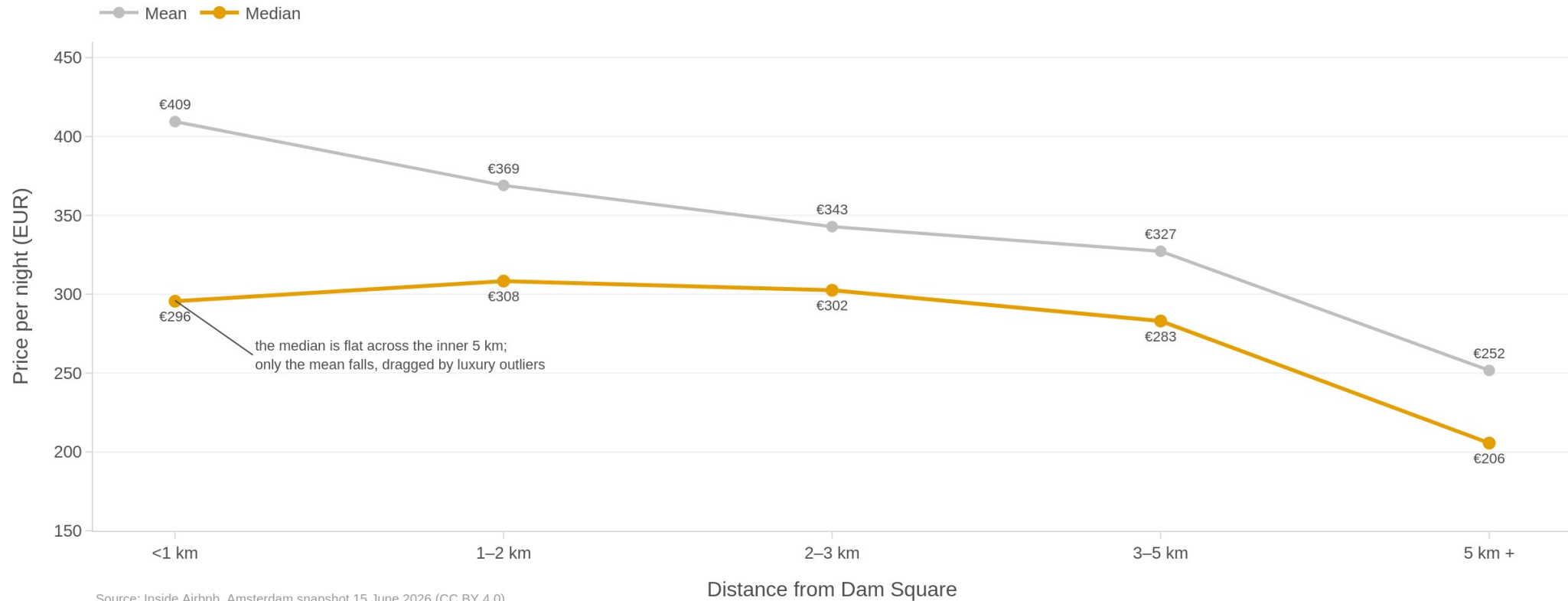
Review scores climb steadily with price — the effect is real but small



Mean rating rises monotonically across every band: 4.62 under €100, then 4.73, 4.79, 4.85, and 4.88 above €300. The direction is consistent, but the whole range spans 0.26 points on a five-point scale where almost everything sits above 4.6. Price predicts quality in the data and barely matters in practice.

QUESTION 9 · WHERE IN THE CITY ARE PRICES HIGHEST?

The famous centre premium is an artefact of the mean — by median it does not exist



By mean, price falls cleanly from €409 within a kilometre of Dam Square to €252 beyond five — the expected gradient. By median, the inner four rings are flat at €284–308, and only the outer ring drops, to €206. The mean gradient is produced by a handful of extreme central listings, up to €11,412 a night. The single most important finding in this deck is a methodological one.

QUESTION 10 · WHERE IN THE CITY ARE PRICES HIGHEST? (SPATIAL VIEW)

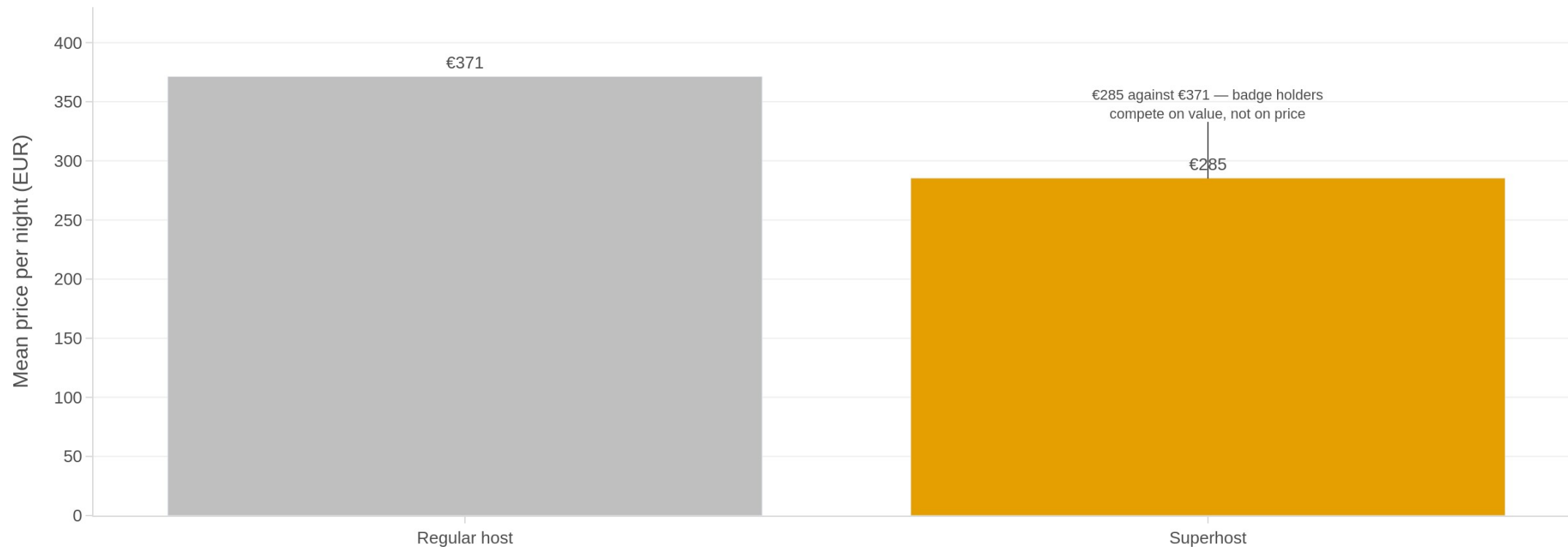
Listings blanket the ring; the highest prices scatter rather than cluster



Plotting all 6,377 listings by coordinate confirms what the gradient chart implies: density is highest in the ring west and south of the centre, and the most expensive listings are scattered individual properties rather than a contiguous premium zone. There is no sharp price boundary anywhere in the city.

QUESTION 11 · DOES SUPERHOST STATUS CARRY A PRICE PREMIUM?

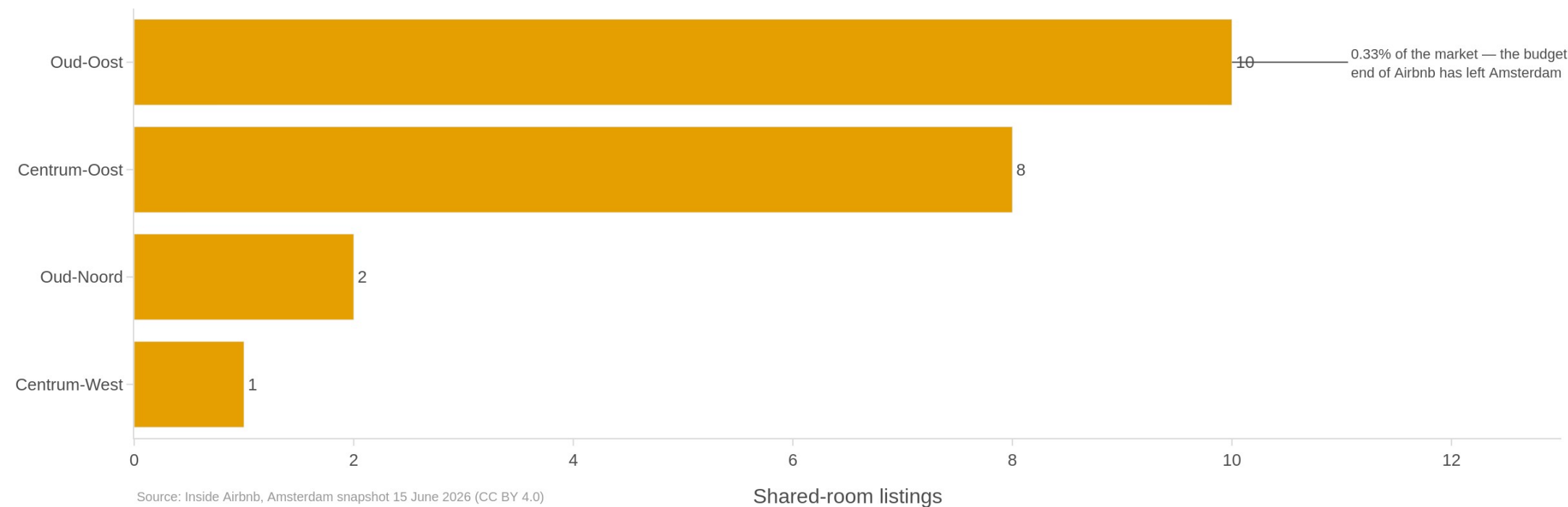
Superhost status carries no price premium — it carries a discount



Superhosts average €285.48 against €371.32 for regular hosts, 23% lower. The badge is earned through response rate, cancellation record and sustained high ratings — all of which reward consistent delivery rather than positioning. These hosts appear to compete on value, and the badge is a quality signal that does not convert into price.

QUESTION 12 · WHERE DOES THE SHARED-ROOM TYPE STILL EXIST?

Shared rooms have all but vanished — 21 listings in a market of 6,377



Only four neighbourhoods have a single shared-room listing between them: Oud-Oost (10), Centrum-Oost (8), Oud-Noord (2) and Centrum-West (1). At 0.33% of the market, the budget end of Airbnb has effectively left Amsterdam — consistent with a market that has professionalised into whole flats and private rooms at €280+ a night.

A curated subset, built for exploration rather than reading

Four tabs share one set of filters — neighbourhood, room type and price band — held in session state so a selection survives every tab switch.

REPLACE THIS BOX WITH YOUR SCREENSHOT

Capture the first tab with the neighbourhood chart and the price map side by side.

Live dashboard: [paste your Streamlit Community Cloud URL]

Source repo: [paste your GitHub URL]

Filters that change the answer, not just the view

Each tab recomputes its statistics against the current selection: the host comparison, the availability distribution and the seasonal curve all move with the filters rather than restating a fixed chart.

REPLACE THIS BOX WITH YOUR SCREENSHOT

Capture a tab mid-interaction — for example Hosts with two neighbourhoods selected.

Live dashboard: [paste your Streamlit Community Cloud URL]

Source repo: [paste your GitHub URL]

WHAT THE TEN QUESTIONS ADD UP TO

A professionalised market, a soft-touch regulation, and a premium that isn't there

The market has professionalised out of its spare-room origins

Shared rooms are 0.33% of listings (Q12), whole flats dominate the residential ring rather than the tourist centre (Q3), and multi-listing operators hold a large share of inventory (Q4). This is a small-hotel sector, not a home-sharing one.

The regulation shapes behaviour at the edges, not the aggregate

23% of hosts hold their calendars under 30 days (Q2), yet citywide demand has recovered to 1.9× its 2019 level (Q5) across a seven-month season (Q6). The cap constrains individual listings without constraining the market.

The signals people trust do not price the way they expect

Superhosts charge 23% less (Q11), professional hosts 27% less (Q4), review scores span 0.26 points across the whole price range (Q8), and the central location premium disappears under the median (Q9).

The methodological finding outranks all of them: the headline claim this analysis started with — that the centre commands a premium — survives only as long as you use the mean.

LIMITATIONS & NEXT STEPS

What this analysis cannot tell you

Listed price is not paid price	Inside Airbnb records the advertised nightly rate on the scrape date. It excludes cleaning fees, service fees, discounts and seasonal pricing, so every euro figure here is a list price, not revenue.
Availability is not occupancy	availability_365 counts bookable days, not nights sold. Q2 therefore evidences hosts' stated intent around the 30-night cap, not their actual compliance with it.
Reviews under-count stays	Only a fraction of guests review, and the rate varies by listing type and over time. Q5 and Q6 track a proxy whose relationship to real bookings is stable enough for shape, not for level.
Comparisons are uncontrolled	Q4 and Q11 compare group means without adjusting for room type, capacity or location. Both differences are large and consistent, but neither is a causal claim.
A single snapshot	One scrape, 15 June 2026. Listings that were removed or dormant that day are absent, which is most of the 38% dropped in cleaning.

What would strengthen it

Inside Airbnb publishes quarterly snapshots. Stacking several would turn every cross-sectional finding here into a trend — most valuably, whether the share of hosts capping their calendars below 30 days is rising as enforcement tightens.